

ToricGT step-local persistent topology

Each reasoning step/window forms a radius-parametrized directed complex with HDBSCAN stability.

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graph-of-thought trajectory windows

$$h_0 \rightarrow h_1 \rightarrow \dots \rightarrow h_T$$
 ρ_1 ρ_2 ρ_3

density gate

$$d_{mr} = \max(c_i, c_j, d_{ij})$$

persistent neighbors

→ receive pullback $K(p_2) \hookrightarrow K(p_3)$

logged invariants

Betti-0 / cycles

boundary residual

Dirichlet energy

directed holonomy